

Introduction to Machine Learning (25737-2)

Problem Set 4

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1 Divergence minimization

For convenience, we will denote $p_\theta(x)$ to be the distribution whose samples are generated by first sampling $z \sim \mathcal{N}(0, I)$ and then returning the sample $G_\theta(z)$. With this notation, we may compactly express the discriminator's loss as

$$L_D(\phi; \theta) = -\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D_\phi(x)] - \mathbb{E}_{x \sim p_\theta(x)}[\log(1 - D_\phi(x))].$$

1. Show that L_D is minimized when $D_\phi = D^*$, where

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_\theta(x) + p_{\text{data}}(x)}.$$

(Hint: for a fixed x , what t minimizes $f(t) = -p_{\text{data}}(x) \log t - p_\theta(x) \log(1 - t)$?)

2. Recall that $D_\phi(x) = \sigma(h_\phi(x))$. Show that the logits $h_\phi(x)$ of the discriminator estimate the log of the likelihood ratio of x under the true distribution compared to the model's distribution; that is, show that if $D_\phi = D^*$, then

$$h_\phi(x) = \log \frac{p_{\text{data}}(x)}{p_\theta(x)}.$$

3. Consider a generator loss defined by the sum of the minimax loss and the non-saturating loss (these terms are combined into a single loss function for the generator.),

$$L_G(\theta; \phi) = \mathbb{E}_{x \sim p_\theta(x)}[\log(1 - D_\phi(x))] - \mathbb{E}_{x \sim p_\theta(x)}[\log D_\phi(x)].$$

Show that if $D_\phi = D^*$, then

$$L_G(\theta; \phi) = \text{KL}(p_\theta(x) \parallel p_{\text{data}}(x)).$$

4. Recall that when training VAEs, we minimize the negative ELBO, an upper bound to the negative log likelihood.
 - Show that the negative log likelihood, $-\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_\theta(x)]$, can be written as a KL divergence plus an additional term that is constant with respect to θ .
 - Does this mean that a VAE decoder trained with ELBO and a GAN generator trained with the L_G defined in the previous part are implicitly learning the same objective? Explain.

2 Distinguishing Real Data from Generated Data

In this question, we aim to establish a connection between the problem of "distinguishing real data from generated data" and the problem of the "distribution distance between real images and generated images." We denote the distribution of real images by P_d and the distribution of generated images by P_g . We also define a distribution P , which gives us random data, as follows:

$$P(x) = 0.5P_g(x) + 0.5P_d(x)$$

Thus, each data point comes equally likely from one of the two distributions P_g or P_d . Now, we form a set of labeled samples (x, y) in the following way:

- Sample x from the distribution P .
- If x comes from the real distribution P_d , set its label $y = 1$; otherwise (if it comes from P_g), set $y = -1$.

We consider $\mathcal{D}$ to be the set of all discriminators. Our goal is to find the best discriminator within this set. We assume this set has no limitations and that the discriminators have unlimited capacity. By unlimited capacity, we mean that for any arbitrary function $y(x)$, there exists a discriminator in $\mathcal{D}$ with that behavior. Therefore, if you find the optimal function point for each x , you can be sure that a discriminator with such outputs exists. The best discriminator D in the set minimizes the following cost function:

$$R_l(D) = E_{(x,y) \sim P}[l(yD(x))]$$

Here, the cost is defined as the expectation over the data x, y . We now define the optimal value of this function:

$$R_l^*(\mathcal{D}) = \inf_{D \in \mathcal{D}} R_l(D)$$

1. (a) First, we rewrite the expectation in the above relation as an integral over the distribution. Your expression should be similar to the following:

$$\inf_{D \in \mathcal{D}} \left(\int ([...]p_d(x) + [...]p_g(x)) dx \right)$$

- (b) Then, given the unlimited capacity of the set $\mathcal{D}$, we compute the optimal expression using pointwise optimization. Note the effect that unlimited capacity has on the optimization.
- (c) Finally, by substituting the optimum and rearranging terms, show that this value is equal to:

$$-\frac{1}{2} \int f \left(\frac{p_d(x)}{p_g(x)} \right) p_g(x) dx$$

Note that we must obtain the function f . After arriving at this relation, verify that the obtained function f is convex and decreasing with respect to its input. Thereafter, we express the above expression using the concept of f -divergence as:

$$-\frac{1}{2} \mathbb{I}(\mathbb{P}_d, \mathbb{P}_g)$$

2. * In the previous part, we showed that this problem is equivalent to minimizing a divergence between two distributions. Now we want to investigate what types of divergences result from choosing different loss functions $l(x)$.

For each loss function:

- (a) $l_1(x) = \mathbb{I}(x \leq 0)$
- (b) $l_2(x) = \log(1 + e^{-x})$

First, check if the corresponding function f is convex and monotonic. Then extract the optimal discriminator, the function f , and the corresponding divergence.

Hint: the corresponding divergences for the loss functions are defined below. You can use this part to check the correctness of your solution.

- For the Total Variation (TV) divergence: $R(D) = \frac{1}{2}(1 - \mathbb{I}_{\text{TV}}(P_d, P_g))$
 - For the Jensen-Shannon (JS) divergence: $R(D) = \log 2 - \mathbb{I}_{\text{JS}}(P_d, P_g)$
3. * Consider the same mixture setup as before, but now interpret the label as a fair binary random variable $Z \in \{0, 1\}$ that is independent of everything else:
- $Z = 1$ with probability $1/2 \rightarrow$ sample $x \sim P_d$ (real),
 - $Z = 0$ with probability $1/2 \rightarrow$ sample $x \sim P_g$ (generated).

The marginal distribution of X is the 50/50 mixture

$$M(x) \triangleq \frac{P_d(x) + P_g(x)}{2}.$$

The joint distribution of (X, Z) therefore factorises with this marginal on X and uniform $P(Z = 0) = P(Z = 1) = 1/2$.

Prove, step by step, that the mutual information between the sample and its origin equals the Jensen-Shannon divergence:

$$I(X; Z) = \text{JSD}(P_d \| P_g),$$

3 Mode collapse and Wasserstein GAN

One of the main practical challenges in training Generative Adversarial Networks (GANs) is a phenomenon called *mode collapse*.

1. Explain what mode collapse is and how it arises during GAN training.
2. Why is mode collapse problematic for practical applications of GANs? In what kinds of applications is it less of an issue?
3. One of the approaches proposed to mitigate mode collapse is the Wasserstein GAN (W-GAN), where we minimize $W(p_{\text{data}}, p_G)$ instead of minimizing a KL-based loss. The Wasserstein-1 distance between two distributions p and q can be written in its primal (optimal transport) form as

$$W(p, q) = \inf_{\gamma \in \Pi(p, q)} \mathbb{E}_{(x, y) \sim \gamma} [\|x - y\|],$$

where $\Pi(p, q)$ is the set of all couplings with marginals p and q , and $\gamma(x, y)$ indicates how much “mass” must be transported from x to y . By Kantorovich–Rubinstein duality, it also admits the dual form

$$W(p, q) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\mathbb{E}_{x \sim p}[f(x)] - \mathbb{E}_{x \sim q}[f(x)] \right),$$

where the supremum is over all 1-Lipschitz functions f .

Why is the Wasserstein distance used in W-GAN a better choice than the KL divergence for measuring similarity between distributions when training GANs? What fundamental differences between these two distance/divergence measures make the Wasserstein distance more suitable for GAN training?

(Hint: Consider two simple 1D distributions where the supports are disjoint. Compute $\text{KL}(q\|p)$ and $W(p, q)$ and compare their behavior as the generator distribution moves closer to the data distribution.)

4 f-GAN Objective and Hellinger Distance

In the generalized f-GAN method, the following objective function is used:

$$F(\theta, \omega) = \min_{\theta} \max_{\omega} \left\{ \mathbb{E}_{x \sim P} [g_f(V_{\omega}(x))] - \mathbb{E}_{x \sim Q_{\theta}} [f^*(g_f(V_{\omega}(x)))] \right\},$$

where

- P is the true distribution and Q_{θ} is the generated (fake) distribution;
- $V_{\omega}(x)$ is the last-layer output just before the output activation function of the discriminator network;
- the activation function is the sigmoid;
- $g_f(v) = 1 - \exp(-v)$ and $f^*(t) = \frac{t}{1-t}$ for $t < 1$.

1. Write the objective function in the simplest possible form in terms of $p(x)$, $q(x)$ and $D(x)$.
2. For what choice of $D(x)$ is the objective in part 1 optimized (in the simplest possible form)?
3. Using the optimal value of $D(x)$, write $F(\theta, \omega)$ in the simplest possible form and justify your answer.
4. In the case of exact and optimal training of the generator network, what will be the value of $D(x)$?

5 Conditional GAN with Auxiliary Classifier (AC-GAN)

In a conditional GAN, we wish to generate images conditioned on a class label $c \in \{1, \dots, C\}$. We assume we have labeled data $(x, c) \sim p_{\text{data}}(x, c)$, and a prior over classes $p(c)$ (typically uniform or equal to the empirical class frequencies).

The generator takes as input a noise vector $z \sim p(z)$ (e.g. $z \sim \mathcal{N}(0, I)$) and a class label c , and outputs an image

$$x = G_{\theta}(z, c), \quad x \sim p_G(x | c),$$

so that the joint model distribution is

$$p_G(x, c) = p(c) p_G(x | c).$$

The discriminator has two outputs: an adversarial head $D_{\text{adv}}(x) \in [0, 1]$ that predicts whether x is real or fake, and a class head $D_{\text{class}}(c | x)$ that outputs a categorical distribution over class labels (via a softmax).

We define the following losses:

$$L_{\text{adv}}(D) = -\mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D_{\text{adv}}(x)] - \mathbb{E}_{z \sim p(z), c \sim p(c)} [\log (1 - D_{\text{adv}}(G_{\theta}(z, c)))],$$

$$L_{\text{adv}}(G) = -\mathbb{E}_{z \sim p(z), c \sim p(c)} [\log D_{\text{adv}}(G_{\theta}(z, c))],$$

$$L_{\text{class}}(D) = -\mathbb{E}_{(x, c) \sim p_{\text{data}}(x, c)} [\log D_{\text{class}}(c | x)],$$

$$L_{\text{class}}(G) = -\mathbb{E}_{z \sim p(z), c \sim p(c)} [\log D_{\text{class}}(c | G_{\theta}(z, c))].$$

The total generator and discriminator losses are

$$L(G) = L_{\text{adv}}(G) + \lambda L_{\text{class}}(G), \quad L(D) = L_{\text{adv}}(D) + \lambda L_{\text{class}}(D),$$

where $\lambda > 0$ is a trade-off hyperparameter.

1. **What distribution is the generator learning?** Assume that after training, for any class label c and noise vector $z \sim p(z)$, the generator $G_{\theta}(z, c)$ produces realistic images of that class. Specify precisely: (i) which conditional distribution $p_G(x | c)$ is trying to match, and (ii) which joint distribution over (x, c) the model is approximating (under the assumption $p(c) = p_{\text{data}}(c)$).
2. **Interpretation of the four loss terms.** Briefly explain what each of the four losses $L_{\text{adv}}(D)$, $L_{\text{adv}}(G)$, $L_{\text{class}}(D)$, and $L_{\text{class}}(G)$ is optimizing and how it shapes the behavior of the discriminator or generator, respectively. Then, using this interpretation, explain in a few sentences how the choice of λ affects the generator: what happens when λ is very small versus very large, in terms of realism and label consistency of the samples.
3. **Impact on mode collapse.** Compare AC-GAN to an unconditional GAN trained only with L_{adv} . Explain how the classification losses $L_{\text{class}}(D)$ and $L_{\text{class}}(G)$ help reduce certain types of mode collapse, and clarify which kinds of mode collapse are still possible in AC-GAN (e.g. across classes vs. within a fixed class).
4. *** Information-theoretic view of the classification loss.** Assume that for a fixed generator G , the classifier head is Bayes-optimal for the current joint model distribution $p_G(x, c) = p(c) p_G(x | c)$, i.e.

$$D_{\text{class}}(c | x) = p_G(c | x).$$

- (a) Show that, up to an additive constant independent of θ , the generator's classification loss can be written as

$$L_{\text{class}}(G) = H_G(c | x),$$

the conditional entropy of the label given the generated image, where the expectation is taken over $(x, c) \sim p_G(x, c)$.

- (b) Argue that minimizing $L_{\text{class}}(G)$ is equivalent to maximizing the mutual information $I_G(c; x)$ between the label and the generated image (up to an additive constant that does not depend on θ).